# Modelling With Algorithms PRACTICE PAPER 1

# AS-LEVEL

**The total for this paper is 60 marks**

**Advice:**

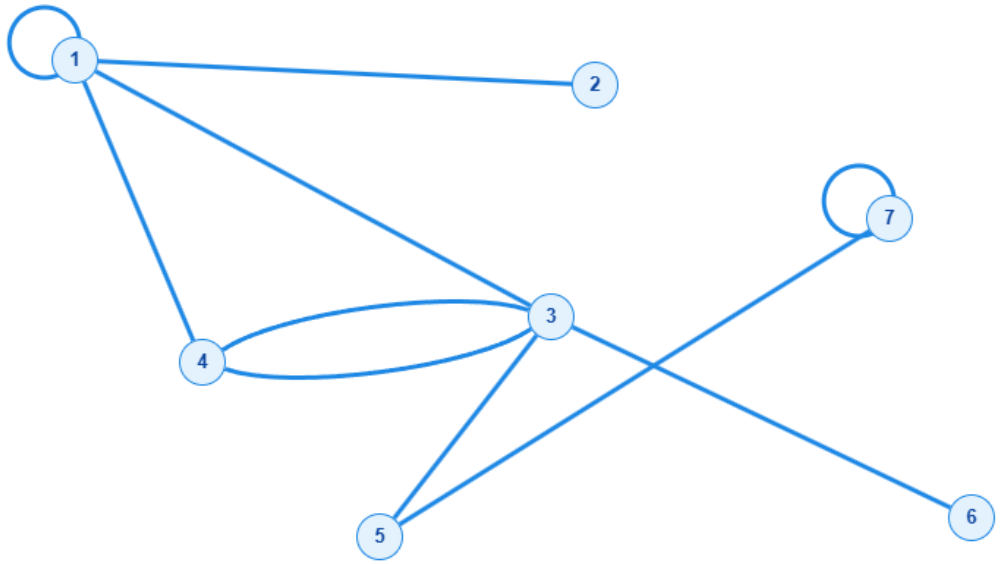
- **Read every question carefully, highlight/underline key information.**
- **Plan a route to the answer, don't mindlessly calculate, as you might miss something**
- **Make sure your answer is reasonable, can a cat be 50 meters tall?**
- **Attempt every question, the first few marks might not be as hard to achieve**
- **When you check a question, assume you made a mistake, then try to find it**
- **Finish the paper quickly, but accurately, and then quadruple check it. Your second set of eyes is always better than your first**
- **Be clear with your method, minimise the number of arrows or boxes you use.**

**Mark totals:**

**1 – 5 marks, 2 – 8marks, 3 – 5 marks, 4 – 6 marks, 5 – 12 marks, 6 – 13 marks, 7 – 5 marks,**

**8 – 6 marks**

**1** The following graph is provided

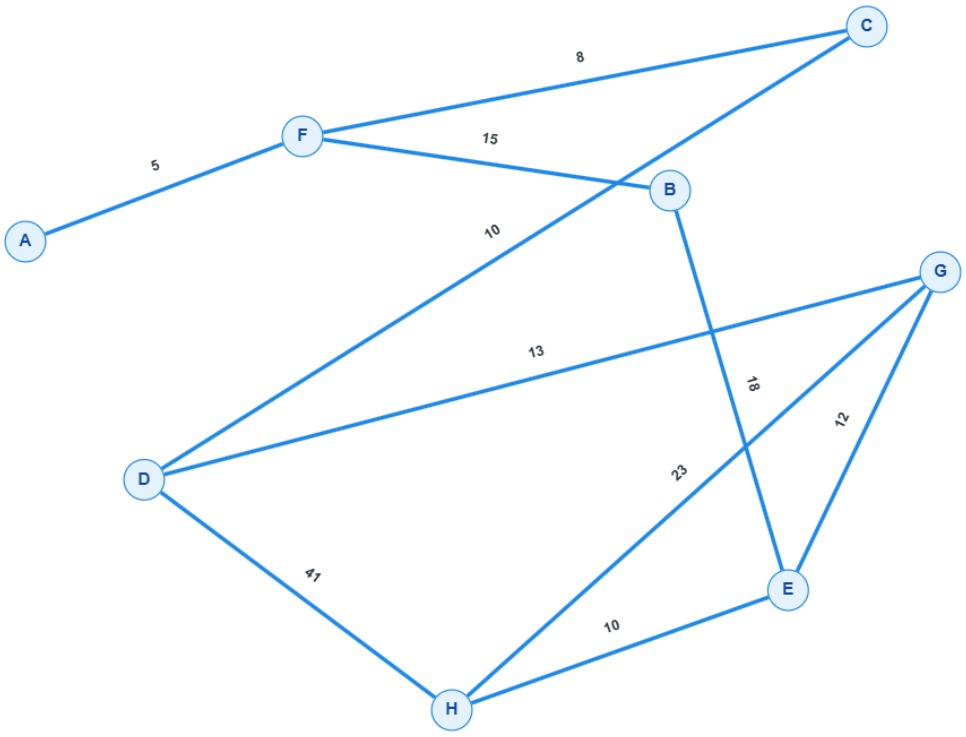



(i) State the nodes with an order of 5 **[1]**

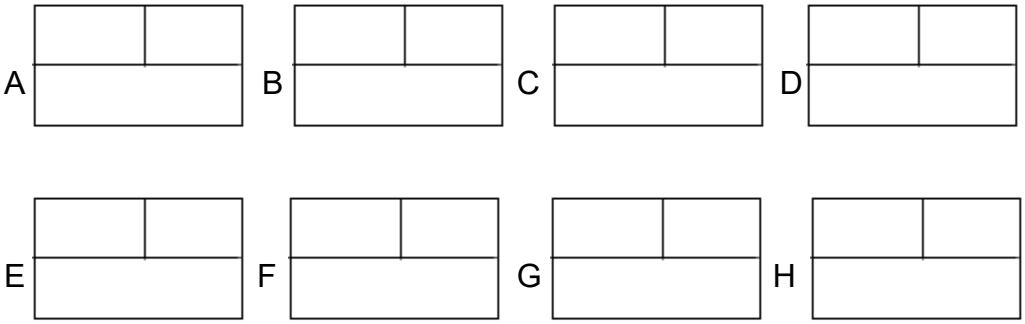
(ii) (A) Form an incidence matrix M to describe the graph **[1]**
(B) Explain what M represents **[1]**

(iii) (A) State, with reason, as to whether this is a connected graph **[1]**
(B) By removing the minimum number of arcs, turn this graph into a simply connected graph **[1]**

**2**




(i) (A) Apply Dijkstra's to a copy of the network, to find the shortest path from A to H, stating the weight of the path. **[5]**




(B) Hence, state the shortest path from C to E, via A, providing its weight **[2]**

(ii) A directed arc is going to be opened from C to G, with a weight of 10, State the new weight of the shortest path from C to E via A **[1]**

**3** Here is an algorithm

```
01 Input X
02 Let N = X/2
03 Let A = N
04 LET N = (A + (X/A))/2
05 IF |A − N| > 0.001 GOTO 03
06 Print N
```

(i) Trace this Algorithm with $X = 2$, recording (to 3 decimal places) each time A, N and X change **[3]**

(ii) A student attempts to trace this algorithm with $X = -2$,
Explain why this is unsuitable **[2]**

**4** A tubing company has received an order for tubes of the following lengths (in cm)

$$13 \;\; 30 \;\; 22 \;\; 26 \;\; 29 \;\; 32 \;\; 59 \;\; 12 \;\; 11 \;\; 38 \;\; 27 \;\; 33$$

They only produce tubes of length 60cm, so they will have to cut tubes from this length, and they cannot weld tubes together to make different lengths from remainders.

(i) Apply the quick-sort algorithm to sort the list of numbers into ascending order. You should use the first value as the pivot for each sublist. **[3]**

(ii) Apply the first fit decreasing algorithm, to show how the tubes can be cut. **[2]**

(iii) Determine whether the packing in part (ii) is optimal **[1]**

**5** A project has the following precedence table

| ACTIVITY | PREDECESSORS | PERIOD (hours) |
|---|---|---|
| A | - | 7 |
| B | - | 5 |
| C | A, B | 4 |
| D | A, B | 6 |
| E | A | 4 |
| F | C, D, E | 4 |
| G | C, D | 6 |
| H | D | 1 |
| I | H | 3 |
| J | H | 3 |

(i) Sketch an activity-on-arc network to represent this table **[3]**

(ii) (A) Complete a forward and backward pass on your network, showing clearly the early start and late finish times for each activity **[3]**
(B) State the minimum completion time for the project **[1]**
(C) State the critical activities for the project **[1]**

For every hour the project is delayed, the contractors face a penalty of $50.
The contractors are currently being paid $1200, to complete the project.

(iii) (A) Calculate the total float for activity H **[2]**

Whilst undergoing the project, the contractors face a delay on H, beginning 5 hours after D finishes.
(B) Calculate the total payment for the contractors. **[2]**

**6** Chris wants to maximise the number of marks he achieves in an Olympiad.
The Olympiad contains many different types of questions: Discrete Maths, Statistics and Core Pure.

Chris has determined that, on average, it takes him 9 minutes per Discrete Maths problem, 6 minutes per Statistics problem and 18 minutes per Core Pure problem. The Olympiad has a total time limit of 1.5 hours.

By analysing past papers, he has found that the average mark for a Discrete Maths problem is 4, a Statistics problem is 3, and a Core Pure problem is 9.

The Olympiad requires the number of Discrete Maths problems solved must at least twice the number of Core Pure problems solved. Furthermore, the number of statistics problems solved must be at least the number of Discrete Maths problems solved.

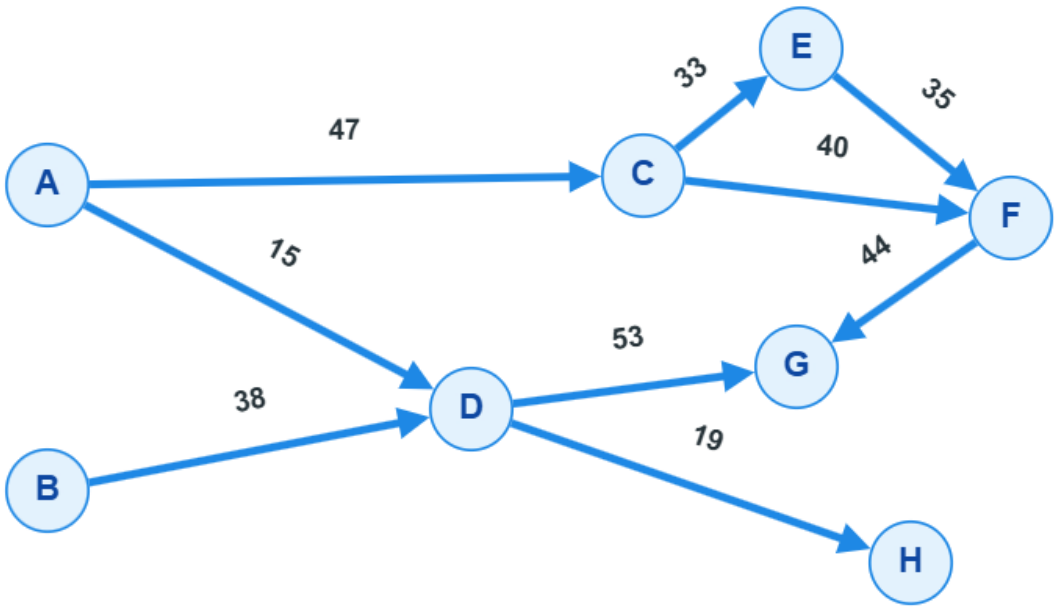
Let x, y, z be the number of Discrete Maths, Statistics and Core Pure problems respectively.

(i) Formulate a simplified integer LP problem to represent this information, showing how each constraint is obtained **[5]**

(ii) (A) Complete an initial Tableau, so that the simplex method may be used to solve the LP. Show how each constraint has been modified using slack variables. **[4]**
(B) Perform one iteration of simplex on your initial tableau **[3]**

(iii) The people who run the Olympiad have now decided that the minimum number of Discrete Maths problems solved must be at least 5.
State the new constraint, and explain why Simplex cannot be used to find a solution to this problem. **[1]**

**7** A plumber uses the following pipe network within a project; she wishes to see if it is suitable. The client requires a minimum flow of 83.




(i) (A) Add a Supersource, S, and Supersink, T, to a copy of the network **[1]**

(B) Explain why not all of the arcs flowing into T can be saturated **[1]**

(ii) (A) Calculate the capacity of the cut $\alpha$, with the partitions {S,A,C,E,F} {B,D,G,H,T} **[1]**

(B) Calculate the capacity of the cut $\beta$, with the partitions {S,A,B,C,D,E,G,H} {F,T} **[1]**

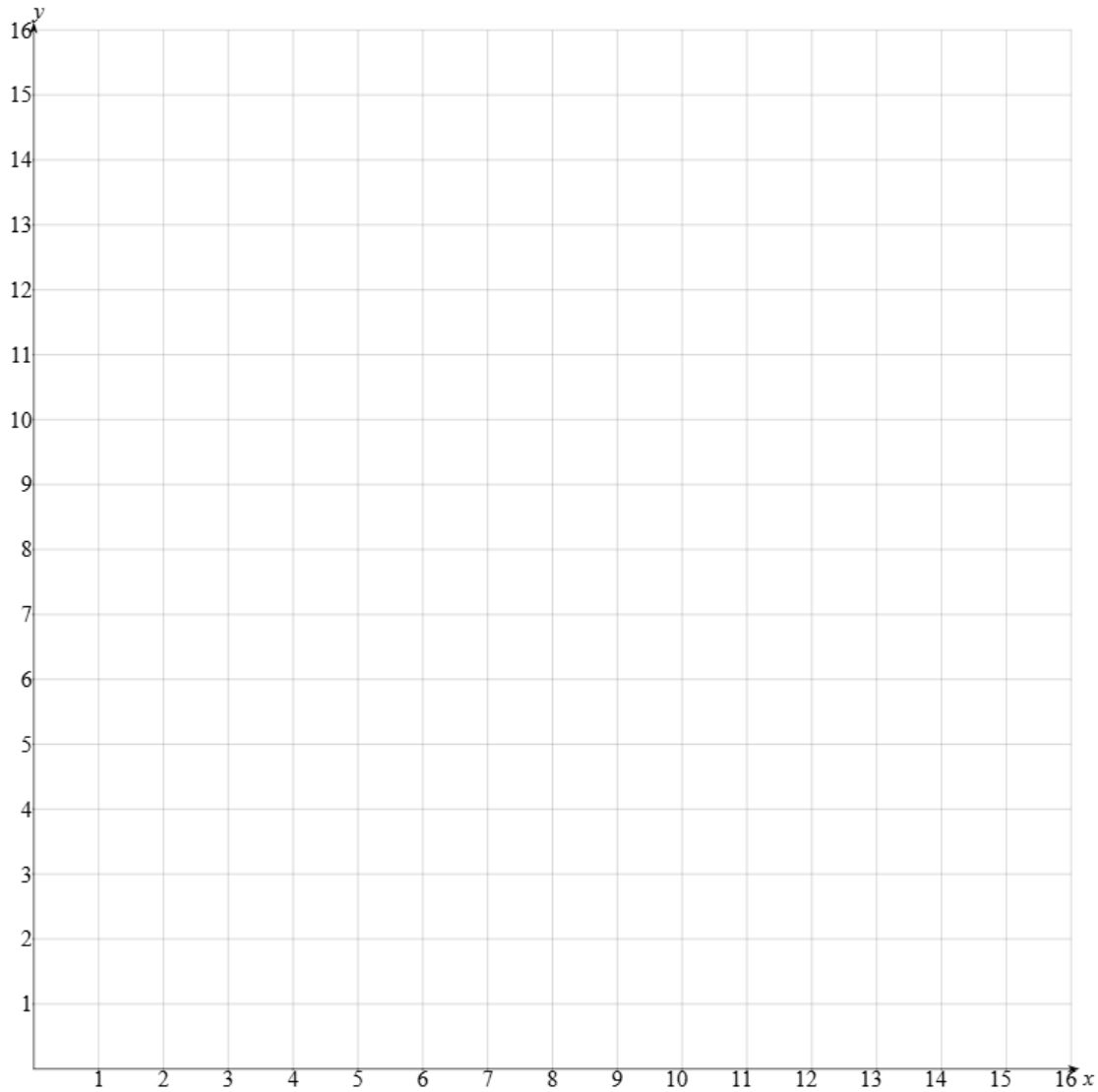
(C) Given that one of $\alpha$ and $\beta$ is the minimum cut,
State, with reason, whether this network is suitable for the client **[1]**

**8** You are given the following linear program.

$Maximise: P = 2x + 4y + 3z$
$Subject\ to: x + y + z = 24$
$2x + 5y \leq 30$
$2x + z \geq 25$
$z \geq 12, x \geq 0, y \geq 0$

(i) By using a suitable substitution, reformulate the constraints in terms of x and y. **[2]**

(ii) Sketch the constraints on the set of axes, clearly indicating the feasible region. **[3]**




(iii) Hence, state the maximum value of P **[1]**